UBCx Phot1x Proposal Draft

Dr. Leonardo Zornberg

***Abstract*—This will be filled out later**

I. Introduction

AN interferometer is a device which can map the relative phase of two different light beams into an amplitude of an outgoing beam, utilizing the principles of constructive and destructive interference. Interferometers may be realized in a variety of different configurations, such as Mach-Zender interferometers (MZIs), Michelson interferometers. In an MZI, which is utilized in this work, light is split from a common source into two beams, which propagate through separate paths, and are then recombined. The relative phase difference between the two paths impacts the amplitude of the recombined beam. This may be used to conduct phase measurements, by relating the measured intensity of the output to determine an unknown phase-difference between the two paths. Alternatively, this phenomenon may be utilized to design power dividers, where a known phase difference between two paths can be used to control the fraction of power coupled into two different output ports. In this work, we explore the use of an MZI to measure not only the group index of an isolated waveguide, but also the group index of coupled waveguides, by measuring the effective Δn_g between two coupled waveguides at fixed L

II. Theory

In this work, we focus on MZIs where the phase difference results from either differences in physical path length for waveguides of equivalent effective index, or in differences effective index for waveguides of equivalent length.

A. Analysis as independent waveguides

1) Y branch splitter

In an ideal y-branch splitter, which splits an input beam of intensity I_{in} into two beams with intensity $I_1 = I_2 = \frac{I_{in}}{2}$. For a Y-branch splitter, the two beams remain in-phase after splitting. Taking the phase at the output port of the splitter as a reference, the electric field at the start of paths 1 and 2 is then given by:

$$E_1^{in} = E_2^{in} = \frac{E_{in}}{\sqrt{2}}, E_{in} \equiv \sqrt{I_{in}}$$

2) Waveguides

As the beams propagate along their respective paths, the phase accumulated due to the waveguide (ϕ_{WG}) is determined by both the effective index (n_{eff}) and pathlength (L)of the waveguide according to:

$$\phi_{WG} = \frac{2\pi}{\lambda_0} n_{eff} * L_{WG}$$

3) Tapers

When the paths are being split or joined, it is necessary to taper the waveguides from the initial/terminal width w_0 with effective index n_0 to the waveguide widths $w_1 \& w_1$ with effective indices $n_1 \& n_2$.

The taper may be approximated as adiabatic if the change in width is sufficiently slow. The criteria for this assumption was derived from Milton and Burns:

$$\theta = \frac{1}{2}\frac{dW}{dz} \leq \frac{\lambda_0}{2 n_{eff} W}$$

If n_{eff} is approximated as a constant, this results in a taper with a parabolic profile

$$W(z) = \sqrt{W_1^2 + 2\frac{\lambda_0}{n_{eff}} z}$$

In this work, where we intentionally utilize n_{eff} which varies with width, we instead relate the width

using the numerically determined relation

$$n_{eff} W dW = \lambda_0 dz \leftrightarrow dW = \frac{\lambda_0 dz}{n_{eff} W}$$

$$\int_{W_0}^{W(z)} n_{eff}(W) W dW = \lambda_0 z$$

Where $n_{eff}(W)$ is approximated by the n_{eff} of a straight waveguide with width W

If the tapering elements are adiabatic (and therefore reflection-free), then the phase contribution to each path may be approximated as

$$\phi_{taper} = \phi_{taper}(W_1, W_2, L) \approx \frac{2\pi}{\lambda_0} \int_0^L n_{eff}\big(W(z)\big) dz$$

For ease of implementation, we may instead use a linear taper satisfying

$$\frac{dW}{dz} = \frac{\lambda_0}{\max\big(n_{eff}\big)(W)} \leq \frac{\lambda_0}{n_{eff} W}$$

In order to balance the phase contribution and lengths of the tapering elements between the two paths of the MZI, we may utilize an indirect tapering approach.

Path 1:

- Adiabatic taper from width $w_0 \to w_1$ with
  - physical length L_{01}
  - phase ϕ_{01}
- Waveguide with width w_1
  - length L_{WG1}
  - phase $\frac{2\pi}{\lambda_0} n_1 * L_{WG1}$
- Adiabatic taper from width $w_1 \to w_2$ with
  - physical length L_{12}
  - phase ϕ_{12}
- Adiabatic taper from width $w_2 \to w_0$ with
  - physical length L_{02}
  - phase ϕ_{02}

This results in a total accumulated phase of

$$\phi_1 = \phi_{01} + \phi_{02} + \phi_{12} + \frac{2\pi}{\lambda_0} n_1 * L_{WG1}$$

And total length

$$L_1 = L_{01} + L_{02} + L_{12} + L_{WG1}$$

Similarly, we may assemble

Path 2:

- Adiabatic taper from width $w_0 \to w_2$ with
  - physical length L_{02}
  - phase ϕ_{02}
- Waveguide with width w_2
  - length L_{WG2}
  - phase $\frac{2\pi}{\lambda_0} n_1 * L_{WG2}$
- Adiabatic taper from width $w_2 \to w_1$ with
  - physical length L_{12}
  - phase ϕ_{12}
- Adiabatic taper from width $w_1 \to w_0$ with
  - physical length L_{01}
  - phase ϕ_{01}

This results in a total accumulated phase of

$$\phi_2 = \phi_{01} + \phi_{02} + \phi_{12} + \frac{2\pi}{\lambda_0} n_2 * L_{WG2}$$

And total length

$$L_2 = L_{01} + L_{02} + L_{12} + L_{WG2}$$

4) Output

At the waveguide outputs, the two beams have accumulated phase dependent on their path,

$$E_1^{out} = E_1^{in} e^{i\phi_1}, \qquad E_2^{out} = E_2^{in} e^{i\phi_2}$$

Resulting in a combined beam with electric field given by

$$E_{out} = \frac{E_1^{out} + E_2^{out}}{\sqrt{2}} = \frac{E_{in}}{2}\left(e^{i\phi_1} + e^{i\phi_2}\right)$$

And intensity given by

$$I_{out} = |E_{out}|^2 = \frac{I_{in}}{4}\left|e^{i\phi_1} + e^{i\phi_2}\right|^2 = \frac{I_{in}}{2}(1 + \cos(\Delta\phi))$$

Where $\Delta\phi = \phi_2 - \phi_1 = \frac{2\pi}{\lambda_0}(n_2 L_{WG2} - n_1 L_{WG1})$

We may calculate the free spectral range (FSR) from the

$$\Delta\phi(\lambda_0 + FSR) = \Delta\phi(\lambda_0) - 2\pi$$

For a small perturbation in λ

$$\Delta\phi(\lambda_0) + \frac{d\Delta\phi}{d\lambda} FSR \approx \Delta\phi(\lambda_0) - 2\pi$$

Substituting the equation for $\Delta\phi$:

$$\frac{d\Delta\phi}{d\lambda_0} = \frac{d}{d\lambda_0}\left(\frac{2\pi}{\lambda_0}(n_2 L_{WG2} - n_1 L_{WG1})\right)$$

$$= -\frac{2\pi}{\lambda_0^2}(n_2 L_{WG2} - n_1 L_{WG1}) + \frac{2\pi}{\lambda_0}\left(L_{WG2}\frac{dn_2}{d\lambda_0} - L_{WG1}\frac{dn_1}{d\lambda_0}\right)$$

$$= -\frac{2\pi}{\lambda_0^2}\left(L_{WG2}\left(n_2 - \lambda_0\frac{dn_2}{d\lambda_0}\right) - L_{WG1}\left(n_1 - \lambda_0\frac{dn_1}{d\lambda_0}\right)\right)$$

Substituting the relationship between effective index n and group index :

$$n_g = n - \lambda\frac{dn}{d\lambda}$$

$$\frac{d\Delta\phi}{d\lambda_0} = -\frac{2\pi}{\lambda_0^2}(L_{WG2}n_{g2} - L_{WG1}n_{g1})$$

$$FSR = \frac{\lambda_0^2}{n_{2g}L_{WG2} - n_{1g}L_{WG1}}$$

These relations may be simplified for each of the MZI architectures being considered in this study

5) Paths of equal index and differing length

If the paths have equal indices ($n_1 = n_2 = n$) but differing length ($L_{WG1} \neq L_{WG2}$) then

$$I_{out} = \frac{I_{in}}{2}\left(1 + \cos\left(\frac{2\pi}{\lambda_0}n(L_{WG2} - L_{WG1})\right)\right)$$

$$FSR = \frac{\lambda_0^2}{n_g(L_{WG2} - L_{WG1})}$$

This relationship may be used to extract the group index n_g from the measured FSR and the known pathlength difference.

6) Paths of differing index and equal length

If the paths have different indices ($n_1 \neq n_2$) and equal lengths ($L_{WG1} = L_{WG2} = L_{WG}$) then

$$I_{out} = \frac{I_{in}}{2}\left(1 + \cos\left(\frac{2\pi}{\lambda_0}(n_{g2} - n_{g1})L_{WG}\right)\right)$$

$$FSR = \frac{\lambda_0^2}{(n_{g2} - n_{g1})L_{WG}}$$

This relationship may be used to extract the group index contrast from the known pathlength.

B. Analysis as coupled waveguides

The waveguides with differing index and equal length can alternatively be modeled as an asymmetric "split" waveguide rather than two distinct waveguides. The two paths, then, can be interpreted as two propagating modes with distinct group indices of n_{g1}^*, n_{g2}^*, resulting in a phase difference of:

$$\Delta\phi = \frac{2\pi}{\lambda_0}(n_{g2}^* - n_{g1}^*)L_{WG}$$

And corresponding free spectal range:

$$FSR = \frac{\lambda_0^2}{(n_{g2}^* - n_{g1}^*)L_{WG}}$$

$$n_{g2}^* - n_{g1}^* = n_2^* - \lambda\frac{dn_2^*}{d\lambda} - \left(n_1^* - \lambda\frac{dn_1^*}{d\lambda}\right)$$

$$= n_2^* - n_1^* - \lambda\frac{d(n_2^* - n_1^*)}{d\lambda}$$

$$\Delta n_g^* = \Delta n_{eff}^* - \lambda\frac{d\Delta n_{eff}^*}{d\lambda}$$

We may solve this as a differential equation to evaluate

$$\Delta n_{eff}^* = \lambda\left(\Delta n_0^* - \int_{\lambda_0}^{\lambda}\Delta n_g^*(\lambda')d\lambda'\right)$$

This may be related to the coupling constant of the coupled waveguides, resulting in

$$\kappa^2 = \Delta\beta^{*2} - \Delta\beta^2 = \left(\frac{2\pi}{\lambda}\right)^2\left(\Delta n_{eff}^{*2} - \Delta n_{\mathrm{eff}}^2\right)$$

III. Modeling and Simulation

A. Waveguide selection

In our design, we consider strip waveguides with height fixed at 220nm, and variable width. In order to select waveguide widths of interest, a parameter sweep was performed in Lumerical mode to calculate effective index and group index vs width.

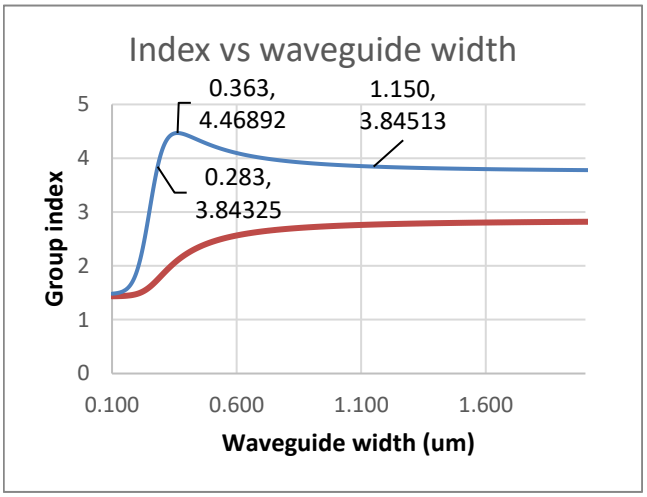


Fig. 1. Plot of group index (blue) and effective index (red) as a function of waveguide width for a silicon strip waveguide with thickness of 220nm. The local maximum in group index occurs at 363nm, and the maximum width considered for this effort is 1.15um

3 waveguide widths of interest were selected:

#	width (um)	ng	neff	Note:
1	0.363	4.46892	4.46892	max ng
2	1.150	3.84513	3.84513	max w
3	0.283	3.84325	3.86360	matched to (2)

The effective index of each waveguide may be expressed as a compact model

$$n_{eff}(\lambda) = c_0(\lambda_0) + c_1(\lambda - \lambda_0) + \cdots + c_n(\lambda - \lambda_0)^n$$

#	width (um)	c_0	c_1
1	0.363	2.10356	-1.50837
2	1.150	2.76805	-0.67960
3	0.281	1.73847	-1.32016

The transfer function of the interferometer vs. wavelength is given by:

$$T_{out}(\lambda) = \frac{1}{2}\left(1 + \cos\left(\frac{2\pi}{\lambda_0}(n_2(\lambda)L_{WG2} - n_1(\lambda)L_{WG1})\right)\right)$$

Where n_2 and n_1 vary with λ according to the derived compact models

Simulations for the mode indices of the coupled waveguide modes are ongoing, but an incomplete table of expected FSR is presented below

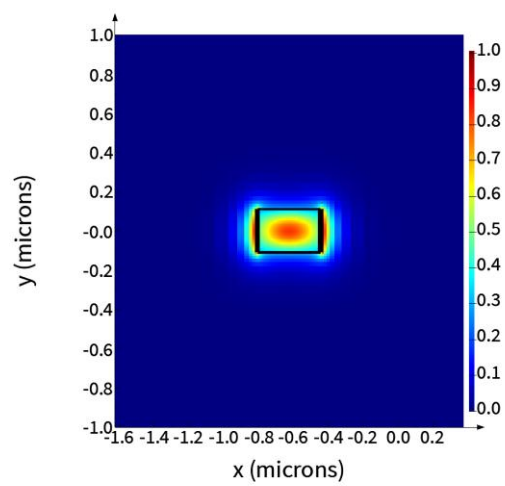
#	w1 (um)	n1	L1 (um)	w2 (um)	n2	L2 (um)	FSR (nm)
1A	0.283	3.86	326	0.283	0.283	100	1.95
2A	0.363	4.47	326	0.363	4.469	100	2.38
3A	0.283	3.86	326	0.283	0.283	100	1.95
4A	0.363	4.47	326	0.363	0.363	100	2.38
5A	1.150	3.85	326	1.150	1.150	100	2.11
6A	0.363	4.47	326	0.363	0.363	100	2.38
7A	1.150	3.85	326	1.150	1.150	100	2.11
8A	0.363	4.47	326	0.363	0.363	100	2.38

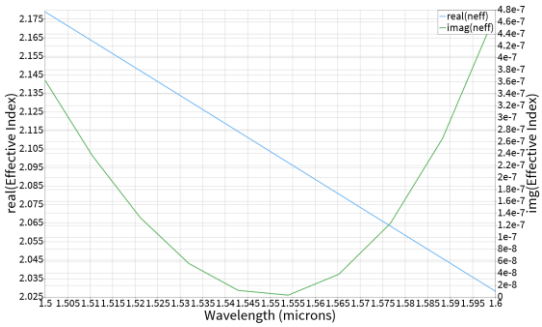
Table of control MZI architectures with fixed waveguide widths and varying waveguide lengths

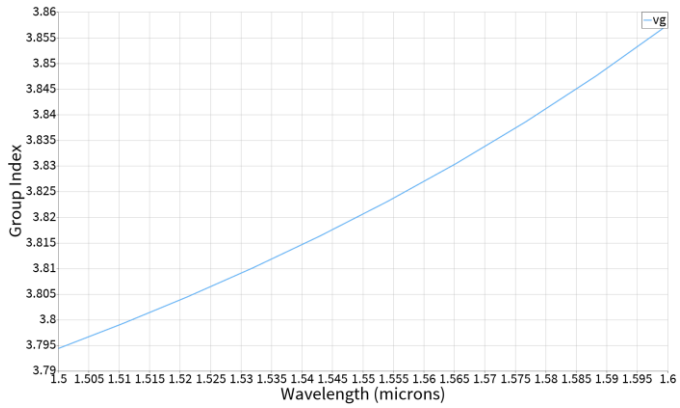
	w1 (um)	n1	L1 (um)	Gap (um)	w2 (um)	n2	L2 (um)	FSR (nm)
1B	0.283	4.01	314	0.1	0.363	4.126	314	66.46
2B	0.283	3.93	314	0.2	0.363	4.302	314	20.80
3B	0.283	3.86	314	0.3	0.363	4.398	314	14.13
4B	0.283	3.86	314	0.4	0.363	4.469	314	12.66
5B	1.150		314	0.1	0.363		314	
6B	1.150		314	0.2	0.363		314	
7B	1.150		314	0.3	0.363		314	
8B	1.150	3.85	314	0.4	0.363	4.469	314	12.28

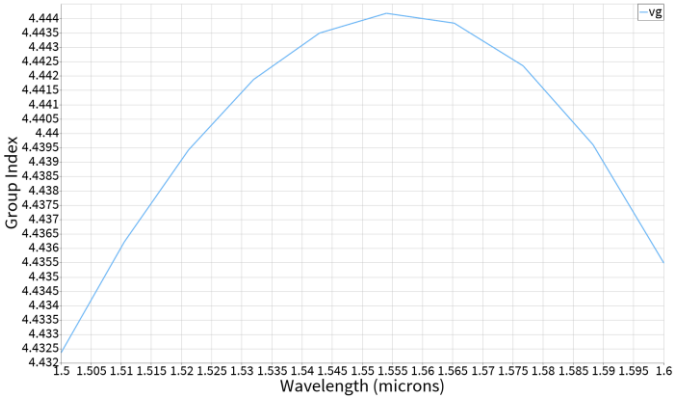
Table of test MZI architectures with fixed waveguide lengths and varying waveguide widths

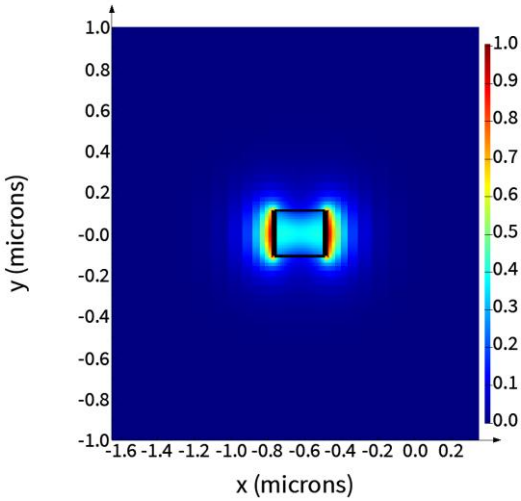
Mode profiles and dispersion curves for the isolated waveguides (363nm, 1150nm, and 283nm) are presented below:

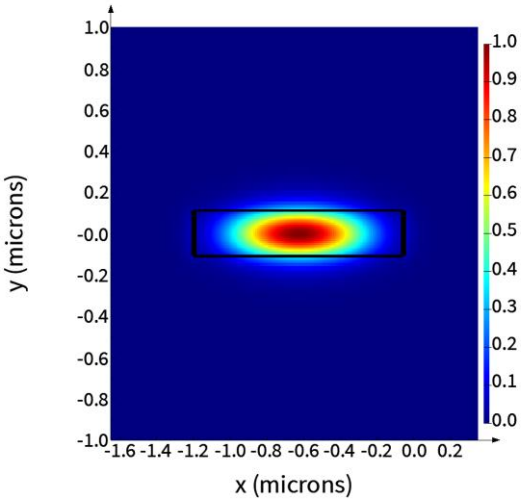


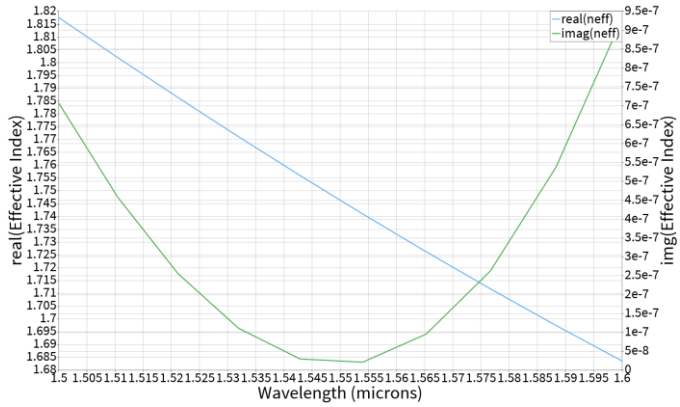

real(neff)
imag(neff)
real(Effective Index)
img(Effective Index)
Wavelength (microns)

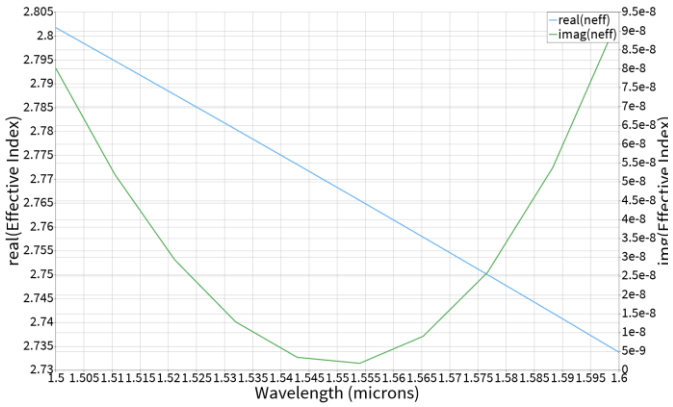

vg
Group Index
Wavelength (microns)

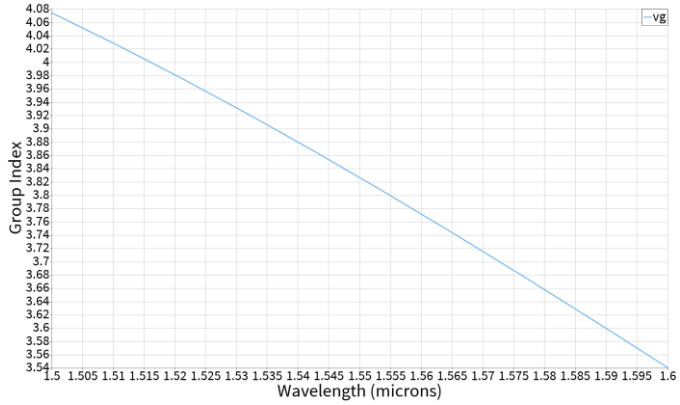

vg
Group Index
Wavelength (microns)


y (microns)
x (microns)

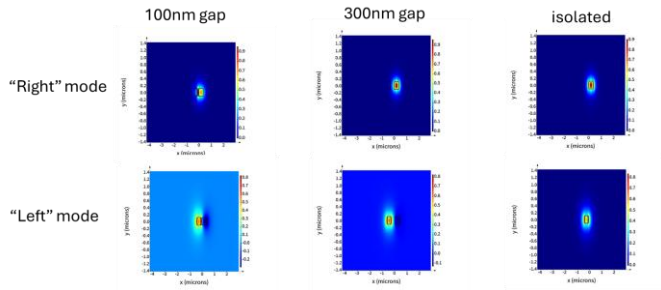

y (microns)
x (microns)


real(neff)
imag(neff)
real(Effective Index)
img(Effective Index)
Wavelength (microns)


real(neff)
imag(neff)
real(Effective Index)
Wavelength (microns)

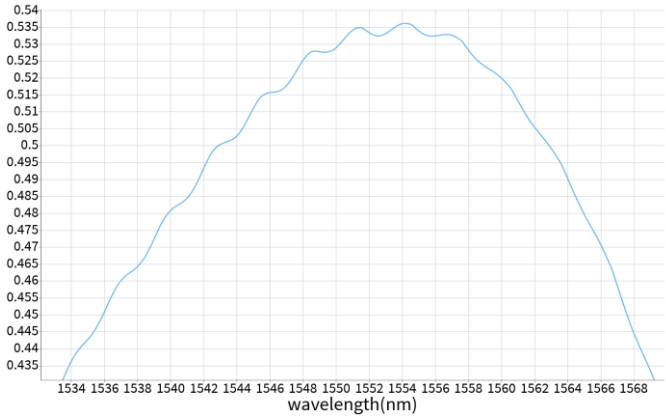

vg
Group Index
Wavelength (microns)

For the analysis of the mode coupling, we may generate comporable field plots



Mode profiles for "left" and "right" modes of the coupled waveguide simulation, as a function of gap size

We may then generate a compact model for each of the left and right mode for each combination of waveguides



Simulation for the coupled modes is ongoing, but an incomplete table of the expected FSR is presented below

Simulation of measured intensity

IV. Fabrication

Lorem ipsum

V. Experimental data

Lorem ipsum

VI. Analysis

Lorem ipsum

VII. Conclusion

Lorem ipsum

References

[1] Zeqin Lu, Han Yun, Yun Wang, Zhitian Chen, Fan Zhang, Nicolas A. F. Jaeger, and Lukas Chrostowski, "Broadband silicon photonic directional coupler using asymmetric-waveguide based phase control," Opt. Express 23, 3795-3808 (2015)